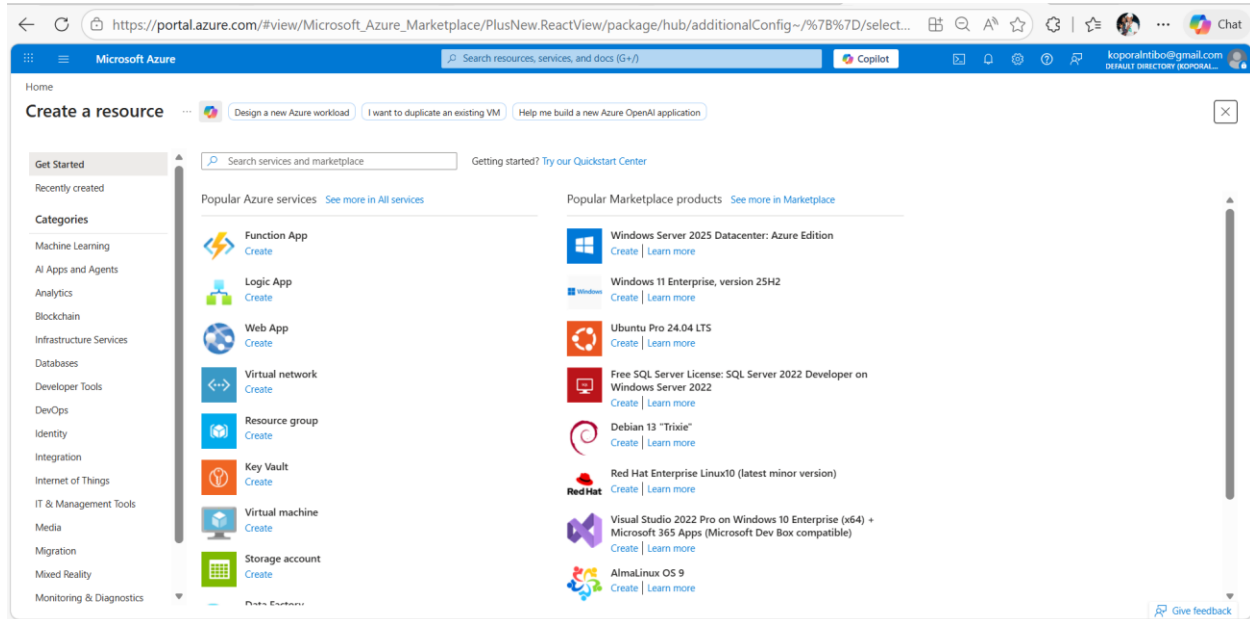


Chapter 2 Lab: Azure Compute Services

Task 1: Create a Linux Virtual Machine (IaaS)



Start by clicking **“Create a resource”** and use the search bar to find **“Virtual Machine”**, then select **“Create.”** This will open the Virtual Machine setup page. From there, choose your subscription (for example, **Azure subscription 1**) and create a new resource group by selecting the **“Create new”** option and giving it an appropriate name.

Next, assign a name to your Virtual Machine, usually based on your surname as instructed. After that, select the region, such as **South Africa North**, to ensure better performance based on your location.

For the image, you can choose **Ubuntu Server 22.04 LTS** as required, but if it is not available, **Ubuntu Server 24.04 LTS** can be used as an alternative. This version provides a secure and stable Linux environment with long-term support and is well optimized for Azure, offering improved performance and faster startup times.

When selecting the size, choose a cost-effective option like **B2ats_v2**, which is suitable for low-demand workloads such as small applications, development environments, or basic web servers.

Then, configure the administrator account by setting the username as **“azureuser”** and creating a strong password to ensure secure access to the virtual machine.

Finally, configure the networking settings to allow **SSH (port 22)** so that you can connect remotely to the Linux machine. Once everything is set, proceed to create the Virtual Machine.

Home

Create a virtual machine

Resource group * RG-ComputeLab [Create new](#)

Instance details

Virtual machine name * VM-Linux-Ntibo ✓

Region * (Africa) South Africa North [Deploy to an Azure Extended Zone](#)

Availability options Availability zone

Zone options

☒ Self-selected zone
Choose up to 3 availability zones, one VM per zone

☐ Azure-selected zone (Preview)
Let Azure assign the best zone for your needs

Availability zone * Zone 1

You can now select multiple zones. Selecting multiple zones will create one VM per zone. [Learn more](#)

Security type Trusted launch virtual machines [Configure security features](#)

Image * Ubuntu Server 24.04 LTS - x64 Gen2 [See all images](#) | [Configure VM generation](#)

< Previous Next : Disks > Review + create

- I used the default **Azure subscription 1** because my account did not have access to the Azure for Students option. This subscription still provided the necessary features to complete the lab tasks.
- I selected the region **South Africa North** since it is closest to my location, making it suitable as a data center for better performance.

Run with Azure Spot discount ☐

Size * Standard_B2ats_v2 - 2 vcpus, 1 GiB memory (US\$8.91) (free services eligible) [See all sizes](#)

Enable Hibernation ☐ Hibernation does not currently support Trusted launch and Confidential virtual machines for Linux images. [Learn more](#)

Administrator account

Authentication type

☒ SSH public key

☐ Password

< Previous Next : Disks > Review + create

Give feedback

Microsoft Azure

Search resources, services, and docs (G+/I)

Copilot

Home

Create a virtual machine

Help me choose the right VM size for my workload | Help me create a VM optimized for high availability | Help me create a low cost VM

enable hibernation ⓘ

ⓘ Hibernation does not currently support Trusted launch and Confidential virtual machines for Linux images. [Learn more](#)

Administrator account

Authentication type ⓘ

☐ SSH public key

☒ Password

Username * ⓘ

Password *

Confirm password *

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports ⓘ

☐ None

☒ Allow selected ports

Select inbound ports *

⚠ This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.

< Previous | Next : Disks > | **Review + create**

Microsoft

Sign out

Estim

Costs ind
dependi
subscrip
exchange
licensing
limited on
preview t

Give

Basic

Virtu

Ima

Ubu

Size

Standard_D8s_v3

\$92.71

Disks

\$0.00

Networking

\$0.00

Management

\$0.00

Estimated monthly cost

\$92.71

Give feedback

After successfully creating and deploying the Virtual Machine, you can use **Windows PowerShell** to manage and access it. PowerShell is useful for automation, efficiency, and seamless integration with Microsoft services.

To connect to the Virtual Machine, open PowerShell and enter the command:
ssh azureuser@PublicIP

In this command, **azureuser** represents the administrator username you created, while **PublicIP** refers to the virtual machine's public IP address, which can be found once the VM is running.

You will then be prompted to enter the administrator password, after which you will gain access to the Virtual Machine's environment.

```
azureuser@VM-Linux-Ntibo: ~
PS C:\WINDOWS\system32> ssh azureuser@102.37.142.9
The authenticity of host '102.37.142.9 (102.37.142.9)' can't be established.
ED25519 key fingerprint is SHA256:zKkNwJP0yNBThSJ3fghDcOpalzkMmShyR47iTo4qvko.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '102.37.142.9' (ED25519) to the list of known hosts.
azureuser@102.37.142.9's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1010-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Apr  2 09:18:49 UTC 2026

System load:  0.02      Processes:      123
Usage of /:   5.7% of 28.02GB    Users logged in:  0
Memory usage: 3%          IPv4 address for eth0: 10.0.0.4
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

azureuser@VM-Linux-Ntibo:~$
```

Task 2: Deploy Azure App Service (PaaS)

https://portal.azure.com/#create/Microsoft.WebSite

Microsoft Azure

Home > Create a resource

Create Web App

App Service Web Apps lets you quickly build, deploy, and scale enterprise-grade web, mobile, and API apps running on any platform. Meet rigorous performance, scalability, security and compliance requirements while using a fully managed platform to perform infrastructure maintenance. [Learn more](#)

Project Details

Select a subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Resource Group * [Create new](#)

Instance Details

Name -euH9hya7gmqehqac.southafricanorth-01.azurewebsites.net

☒ Secure unique default hostname on. [More about this update](#)

Publish * ☒ Code ☐ Container

Runtime stack *

Operating System * ☒ Linux ☐ Windows

Region * Not finding your App Service Plan? Try a different region or select your App Service Plan.

[Review + create](#) [< Previous](#) [Next: Database >](#)

You will be required to deploy a web application without managing servers, using Platform as a Service (PaaS). You can do this by creating a new resource, and searching for App Service in the search bar, then clicking create.

You will be then required to fill in the required details to create the Web App. This includes Project details, where subscriptions and resource groups are needed. Selecting your subscription “Azure for Students” and Resource Group, the one created on the previous task. Instance details will also be required, where naming your Web App using your surname and selecting “Code” as publish option. Code – the standard choice for deploying applications built with specific programming languages and frameworks (such as .NET and Node.js) directly to a fully managed hosting environment. Runtime stack = .NET (LTS), as it ensures maximum stability, security, and performance for production applications.

In my case using .NET 8 (LTS) which provides long-term stability and support, significant performance enhancements, and built-in features optimized for cloud native development, instead of the recommended .NET 6 or Node.js.

Feature / Criteria	.NET 8	.NET 6	Node.js
Performance	Highest throughput, improved JIT, native AOT compilation	Stable, but slower than .NET 8	Fast for I/O-bound tasks, weaker for CPU-heavy
Azure Integration	Deep integration with Azure Functions, App Service, AKS, Cosmos DB	Supported, but lacks latest optimizations	Supported, especially with serverless (Functions)
Microservices	Strong with ASP.NET Core, gRPC, Dapr, Kubernetes	Good, but less optimized for container workloads	Excellent for lightweight, event-driven microservices
Scalability	Enterprise-grade, handles millions of requests with low latency	Solid, but less efficient under extreme load	Scales horizontally well, but requires more tuning
Developer Experience	Modern C# features, better tooling in VS Code & Visual Studio	Mature ecosystem, but fewer new features	Huge npm ecosystem, fast development

Feature / Criteria	.NET 8	.NET 6	Node.js
Best Use Cases	Enterprise APIs, financial systems, cloud-native apps	Legacy modernization, stable enterprise apps	Real-time apps, APIs, streaming services, lightweight backends

The screenshot shows the 'Create Web App' wizard in the Microsoft Azure portal. The configuration is as follows:

- Operating System:** Linux (selected), Windows (unselected)
- Region:** South Africa North
- Pricing plans:**
 - Linux Plan (South Africa North): (New) APS-RGComputeLab-9175
 - Pricing plan: Basic B1 (100 total ACU, 1.75 GB memory, 1 vCPU)
- Zone redundancy:**
 - Enabled: Your App Service plan and the apps in it will be zone redundant. The minimum App Service plan instance count will be two.
 - Disabled: Your App Service plan and the apps in it will not be zone redundant. The minimum App Service plan instance count will be one. (Selected)

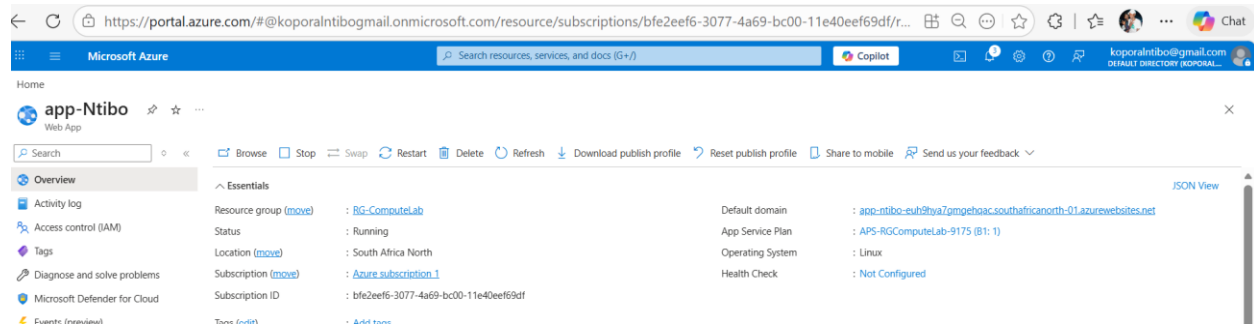
Navigation buttons at the bottom include 'Review + create', '< Previous', and 'Next: Database >'.

App Service Plan Selection

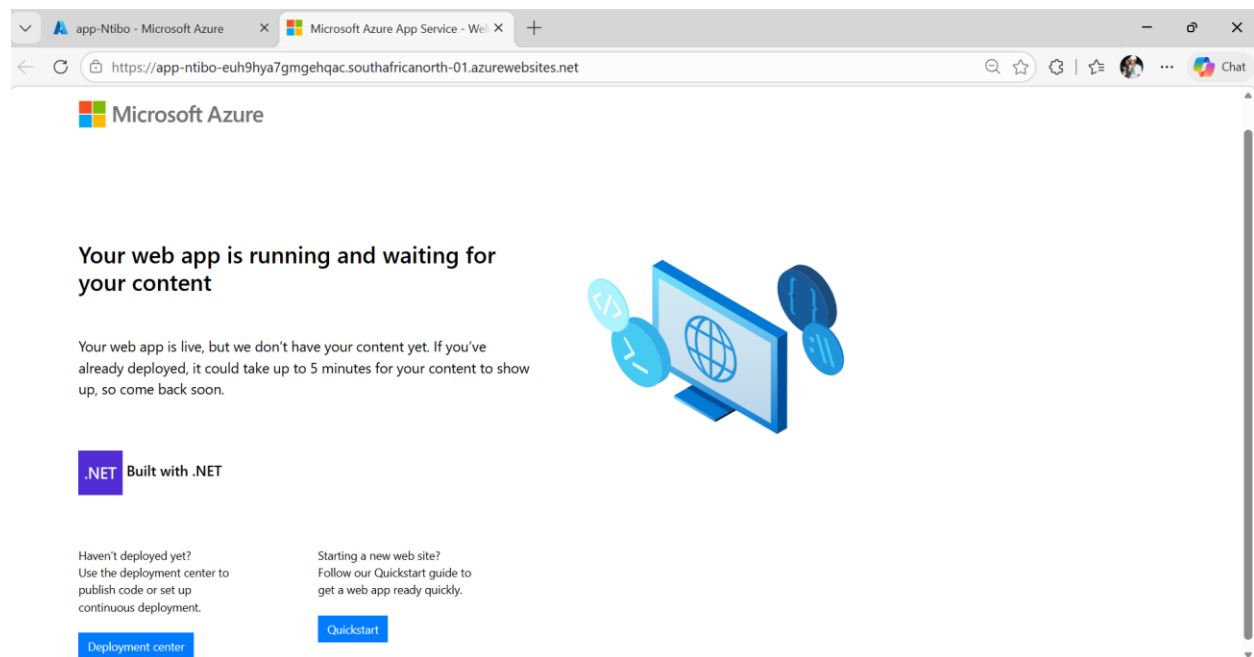
App Service plan pricing determines the location, features, cost, and compute resources associated with an application. In this project, the **Basic B1 plan** was selected as it is a cost-effective, dedicated compute tier suitable for low-traffic applications, as well as development and testing environments.

The same pricing plan was used for deploying the application because the **Free (F1) tier**, which was initially recommended, was not available in the Azure account.

After configuring the App Service plan, the Web App was successfully created and deployed.



Using the Web App's default domain after deployment will allow you to see a default web page of your web app.



Task 3: Create Azure Function (Serverless)

Serverless Computing and Azure Functions

Serverless computing is an execution model in which developers deploy code as functions, while the cloud provider automatically manages the underlying infrastructure. Although the term “serverless” is used, servers are still involved; however, they are abstracted away from the developer, allowing a greater focus on application logic rather than system management.

Azure Functions, a service provided by Microsoft, is a serverless compute solution that enables the execution of event-driven code without the need to manage servers. It automatically scales based on demand, ensuring efficient resource utilization.

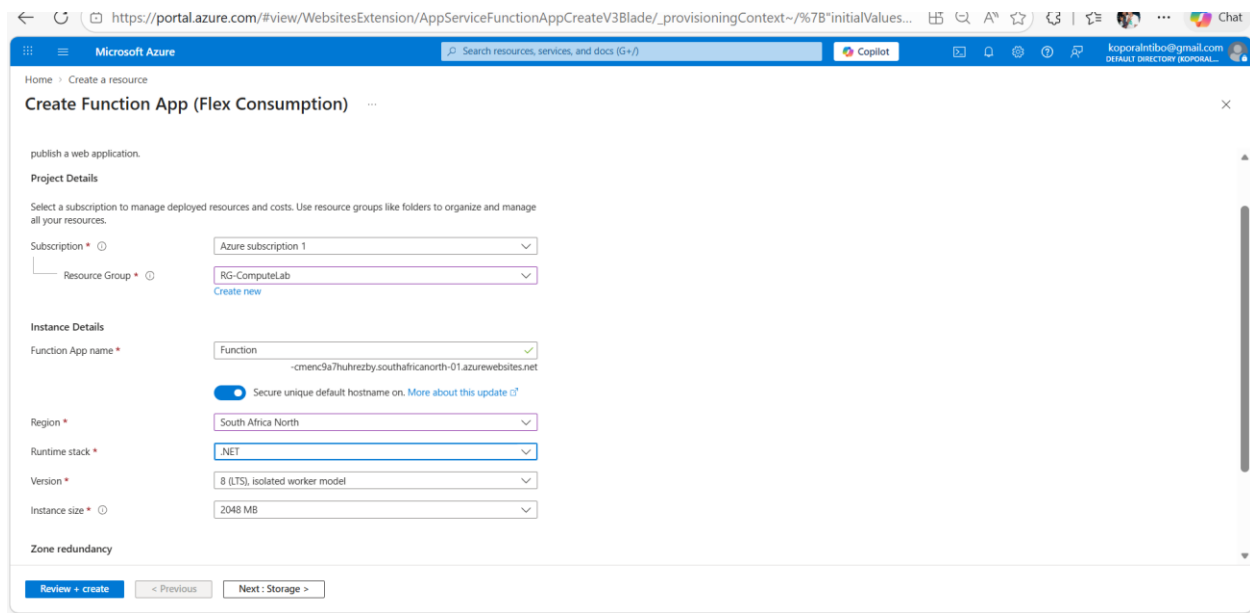
Additionally, it integrates seamlessly with other Azure services and follows a pay-per-execution pricing model, meaning users are only charged for the time their code runs.

This makes Azure Functions particularly suitable for building APIs, automation tasks, and microservices.

The screenshot shows the 'Create Function App' page in the Azure portal. The page title is 'Create Function App' and the URL is 'https://portal.azure.com/#create/Microsoft.FunctionApp'. The page is titled 'Select a hosting option' and includes a link to 'Learn more about Functions hosting options'. Below this, there is a table comparing six hosting plans: Flex Consumption, Functions Premium, App Service, Container Apps environment, and Consumption (Windows). The table lists various features and their availability for each plan, such as 'Scale to zero', 'Scale behavior', 'Virtual networking', 'Dedicated compute and prevent cold start', and 'Max scale out (instances)'. The 'Flex Consumption' plan is selected by default.

Hosting plans	Flex Consumption	Functions Premium	App Service	Container Apps environment	Consumption (Windows)
Scale to zero	✓	-	-	✓	✓
Scale behavior	Fast event-driven	Event-driven	Metrics based	Event-driven with KEDA	Event-driven
Virtual networking	✓	✓	✓	✓	-
Dedicated compute and prevent cold start	Optional with Always Ready	Minimum of 1 instance required	Minimum of 1 instance required	Optional with minimum replicas	-
Max scale out (instances)	1000	100	30	300	200

Select



publish a web application.

Create Function App (Flex Consumption)

Project Details

Select a subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Resource Group *
[Create new](#)

Instance Details

Function App name *
-cmenc9a7uhrezby.southafricanorth-01.azurewebsites.net

☒ Secure unique default hostname on. [More about this update](#)

Region *

Runtime stack *

Version *

Instance size *

Zone redundancy

[Review + create](#) [< Previous](#) [Next: Storage >](#)

NET 8 Configuration and Instance Size

.NET 8 (LTS) provides long-term stability and support, along with significant performance improvements and built-in features that are optimized for cloud-native development. These enhancements make it a suitable choice for building scalable and efficient applications in modern cloud environments.

An instance size of **2048 MB** was selected to support the runtime stack version used during the creation of the function. This ensures adequate memory allocation for smooth execution and improved performance of the application.

Home > Function App > Create Function App

Create Function App (Consumption (Windows))

Project Details

Select a subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ

Azure for Students

Resource Group * ⓘ

RG-ComputeLab

[Create new](#)

Instance Details

Function App name *

Function

-a2f4hjaxbnc6guhw.southafricanorth-01.azurewebsites.net



Secure unique default hostname on. [More about this update](#)

Operating System

Windows

For Linux, [learn more](#)



Flex Consumption is now the recommended serverless hosting plan for Azure Functions. [Learn more.](#)

To create a function for Function App I used Visual Studio Code, because it provides a lightweight, cross-platform environment with the Azure Functions extension, enabling local development, debugging, and seamless deployment to Azure. It supports multiple languages (C#, JavaScript, Python, etc.) and integrates tightly with Azure services, making it ideal for modern serverless workflows.



Azure Functions

C#AzureCloud

Project name

HelloFunction

Location

C:\Users\smmgw\OneDrive\Documents\visual studio\

...

Solution name ⓘ

HelloFunction

☐ Place solution and project in the same directory

Project will be created in "C:\Users\smmgw\OneDrive\Documents\visual studio\HelloFunction\HelloFunction\"

BackNext

Additional information

Azure Functions

C#AzureCloud

Functions worker ⓘ

.NET 8.0 Isolated (Long Term Support)

Function ⓘ

Http trigger

☐ Enlist in Aspire orchestration ⓘ

Aspire version ⓘ

13.1

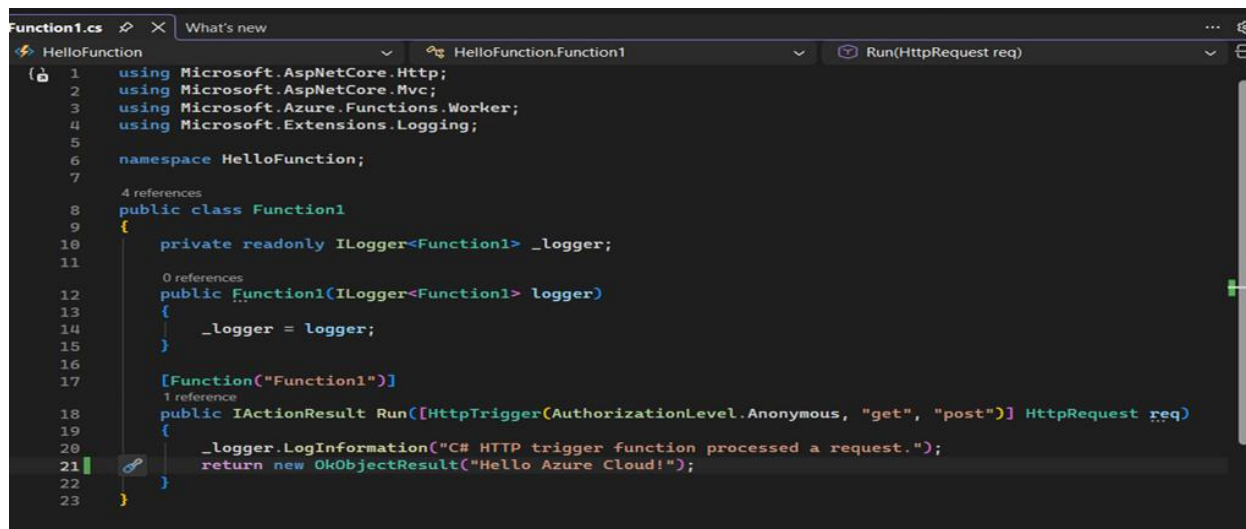
☒ Use Azurite for runtime storage account (AzureWebJobsStorage) ⓘ

☐ Enable container support ⓘ

Authorization level ⓘ

Anonymous

Creating Function in Visual Studio

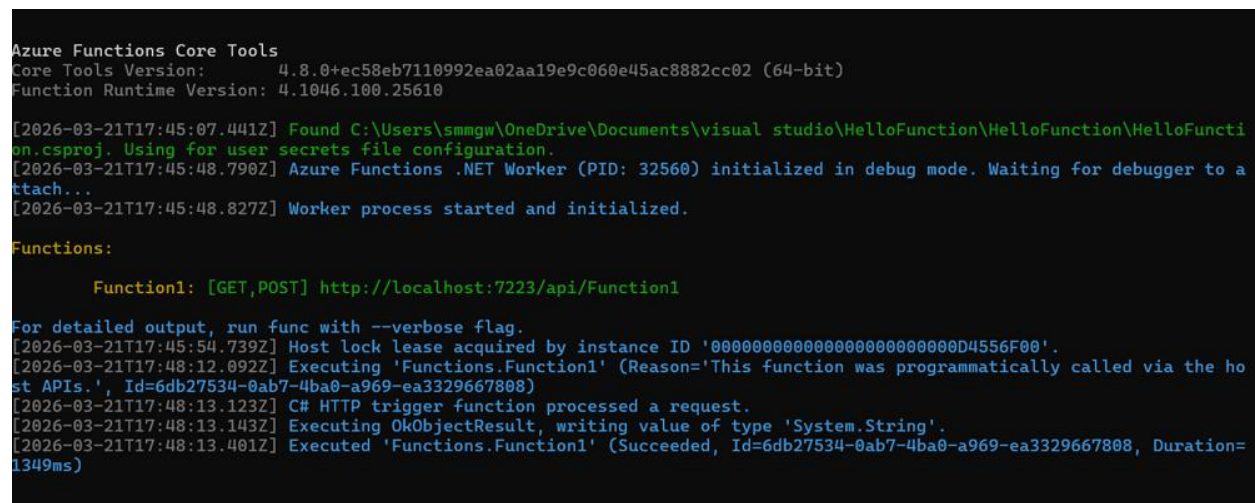


```
Function1.cs
1 using Microsoft.AspNetCore.Http;
2 using Microsoft.AspNetCore.Mvc;
3 using Microsoft.Azure.Functions.Worker;
4 using Microsoft.Extensions.Logging;
5
6 namespace HelloFunction;
7
8 public class Function1
9 {
10     private readonly ILogger<Function1> _logger;
11
12     public Function1(ILogger<Function1> logger)
13     {
14         _logger = logger;
15     }
16
17     [Function("Function1")]
18     public IActionResult Run([HttpTrigger(AuthorizationLevel.Anonymous, "get", "post")] HttpRequest req)
19     {
20         _logger.LogInformation("C# HTTP trigger function processed a request.");
21         return new OkObjectResult("Hello Azure Cloud!");
22     }
23 }
```

Adding Code Inside the Function Editor

After creating the function, code was added inside the function editor to define the logic of the application. The function was configured to return a simple HTTP response message, confirming that the function is running successfully.

This step is important as it allows the developer to test and verify that the Azure Function is correctly deployed and functioning as expected.



```
Azure Functions Core Tools
Core Tools Version: 4.8.0+ec58eb7110992ea02aa19e9c060e45ac8882cc02 (64-bit)
Function Runtime Version: 4.1046.100.25610

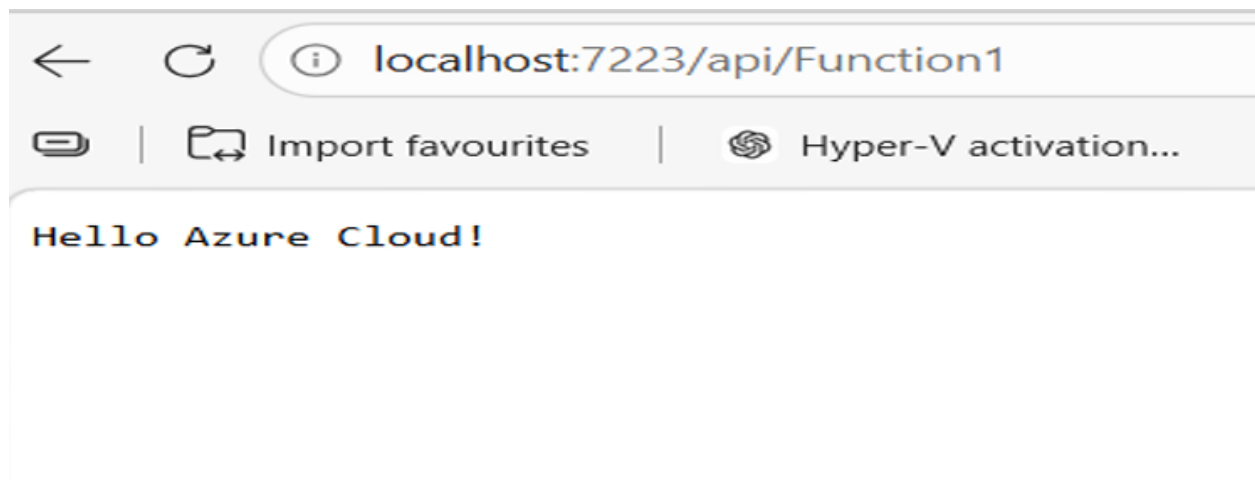
[2026-03-21T17:45:07.441Z] Found C:\Users\smmgw\OneDrive\Documents\visual studio\HelloFunction\HelloFunction\HelloFunction.csproj. Using for user secrets file configuration.
[2026-03-21T17:45:48.790Z] Azure Functions .NET Worker (PID: 32560) initialized in debug mode. Waiting for debugger to attach...
[2026-03-21T17:45:48.827Z] Worker process started and initialized.

Functions:

    Function1: [GET,POST] http://localhost:7223/api/Function1

For detailed output, run func with --verbose flag.
[2026-03-21T17:45:54.739Z] Host lock lease acquired by instance ID '00000000000000000000000000000000D4556F00'.
[2026-03-21T17:48:12.092Z] Executing 'Functions.Function1' (Reason='This function was programmatically called via the host APIs.', Id=6db27534-0ab7-4ba0-a969-ea3329667808)
[2026-03-21T17:48:13.123Z] C# HTTP trigger function processed a request.
[2026-03-21T17:48:13.143Z] Executing OkObjectResult, writing value of type 'System.String'.
[2026-03-21T17:48:13.401Z] Executed 'Functions.Function1' (Succeeded, Id=6db27534-0ab7-4ba0-a969-ea3329667808, Duration=1349ms)
```

Running code in Visual Studio



Testing the Function in a browser (any browser) Azure Function run only when triggered, and they auto-scale automatically.

Comparison: VM (IaaS) vs App Service (PaaS) vs Function (Serverless)

Feature	VM (IaaS)	App Service (PaaS)	Function (Serverless)
OS Control	Yes	No	No
Scaling	Manual / Auto	Auto	Auto
Billing	Per minute	Per hour	Per execution
Best For	Full control apps	Web apps	Event-driven tasks

Task 4: Cleanup

Delete Resources

The selected resources along with their related resources and contents will be permanently deleted. If you are unsure of the selected resource dependencies, navigate to the individual resource page to perform the delete operation. More details of the resource dependencies are available in the manage experience.

1 resources being deleted across 1 subscriptions, 1 resource groups

Name	Resource type	Subscription	Resource Group	
 Function	Function App	Azure for Students	RG-ComputeLab	Remove

Delete confirmation

Deleting the selected resources and their internal data is a permanent action and cannot be undone.

[Delete](#) [Go back](#)

Enter "delete" to confirm deletion *

[Delete](#) [Cancel](#)

Resource Cleanup

Cloud resources incur costs; therefore, deleting them when they are no longer needed helps an organization reduce expenses and optimize the use of cloud storage and computing resources.